IDS 702 Project Proposal

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Project Proposal/Statistical Analysis Plan

Link to your team's Github Repo: https://github.com/chen04301121-cmyk/ids702-ai-unemployment

Dataset Option 1

Project Title: AI Adoption in Job Postings and Unemployment Across Economies

Data citation (use APA formatting or something similar):

International Monetary Fund. (2026). *AI Adoption Index (AIAX)* [Data set]. IMF Data. Retrieved November 4, 2026, from https://data.imf.org/en/datasets/IMF.STA:WPAIAX

International Monetary Fund. (n.d.). *Labor Statistics (LS)* [Data set]. IMF Data. Retrieved November 3, 2026, from https://data.imf.org/en/datasets/IMF.STA:LS

International Monetary Fund. (n.d.). *World Economic Outlook (WEO)* [Data set]. IMF Data. Retrieved November 4, 2026, from https://data.imf.org/en/datasets/IMF.RES:WEO

The AIAX dataset accompanies: Ando, S., Kirti, D., Lin, W., et al. (2026). *How is labor demand changing as AI improves? New global evidence* (IMF Working Paper). International Monetary Fund.

Original data collection date (when was the dataset published?): The AIAX dataset was published by the IMF on November 2, 2026 and contains monthly economy-level indicators for 110 economies from January 2023 to April 2026, derived from online job postings. The Labor Statistics (LS) dataset compiles labor force, employment, and unemployment statistics reported by national statistical organizations; we use the latest vintage of quarterly unemployment rates for 2023 Q1 to 2025 Q4. The WEO dataset provides annual macroeconomic indicators; we use the latest vintage for 2023 to 2025.

Data pull date (when did you download the data you will use?): IMF Labor Statistics (LS): November 3, 2026; IMF AI Adoption Index (AIAX) and World Economic Outlook (WEO): November 4, 2026.

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1 Research Question

This proposal aims to examine the relationship between growing demand for artificial intelligence (AI) skills and unemployment rates across economies.

Primary research question: Among 82 economies observed quarterly from 2023 to 2025, is a higher share of online job postings requiring emerging AI skills associated with the unemployment rate, after adjusting for GDP growth, inflation, GDP per capita, year, and season? Does this association differ between advanced economies and emerging/developing economies?

2 Study Population

The AIAX dataset measures the prevalence of emerging AI skills in online job postings (collected by Lightcast and processed by the IMF) across 110 economies, with monthly economy-level observations. The Labor Statistics (LS) dataset provides labor market indicators, including unemployment rates, that originate from national statistical organizations and are compiled by the IMF. The WEO dataset provides annual macroeconomic indicators (real GDP growth, inflation, and GDP per capita) compiled by IMF.

The unit of observation is an **economy-quarter**, with each observation capturing macroeconomic indicators (e.g. unemployment rate and AI skill demand) for that particular country in a given quarter. We aggregate the monthly AI indicators to quarterly averages and assign annual WEO values to all quarters of the same year. After merging the three sources by ISO country code, year, and quarter, the analytic sample contains 826 economy-quarter observations from 82 economies (41 classified as advanced and 41 as emerging/developing economies by the IMF) between 2023 Q1 and 2025 Q4. Economies without both unemployment and AI data in a given quarter are excluded.

3 Outcomes, Exposures, and Additional Variables of Interest

3.1 Primary Outcome(s)

Variable Name	Description	Coding specification
unemployment_rate	Unemployment rate: percentage of the labor force that is unemployed, by economy and quarter (Labor Statistics dataset)	Continuous, percent (0–100); higher values indicate higher unemployment. Log-transformed in the model.

3.2 Primary Independent Variable(s)

Variable Name	Description	Coding specification
ai_share	Share of online job postings requiring emerging AI skills, all occupations (AIAX "Share of Emerging AI"), quarterly average of monthly values	Continuous, percent; higher values indicate greater demand for AI skills.
economy_group	International Monetary fund (IMF) economy classification (interaction variable)	Categorical: 1 = Advanced economy, 0 = Emerging/developing economy (reference)

3.3 Additional Variables of Interest (including potential confounders)

Variable Name	Description	Coding specification
gdp_growth	Real GDP growth, constant prices (WEO)	Continuous, annual percent change
inflation	CPI inflation, period average (WEO)	Continuous, annual percent change
gdp_per_capita	GDP per capita, current prices (WEO)	Continuous, U.S. dollars; log-transformed in the model
year	Calendar year	Categorical: 2023 (reference), 2024, 2025
qtr	Calendar quarter, to account for seasonality	Categorical: 1 (reference), 2, 3, 4
iso3 / COUNTRY	Economy identifier	Categorical, ISO 3-letter country code; used for the random intercept

4 Statistical Analysis Plan

4.1 Exploratory Data Analysis

The outcome (quarterly unemployment rate) is continuous, so we will summarize it with descriptive statistics and a histogram to assess its distribution and identify extreme values. Because our primary independent variable (the share of job postings requiring emerging AI skills) is also continuous, we will use a scatter plot of unemployment against AI adoption, with points colored and separate linear fits drawn by economy group (advanced vs. emerging/developing), to explore both the main relationship and the planned interaction.

4.2 Statistical Model

We will fit a linear mixed-effects model with a random intercept for each economy:

$$\begin{aligned}\log(\text{Unemp}_{it}) = \beta_0 &+ \beta_1 \text{AI}_{it} + \beta_2 \text{Adv}_i + \beta_3 \left[\text{AI}_{it} \times \text{Adv}_i\right] \\ &+ \beta_4 \text{GDPgrowth}_{it} + \beta_5 \text{Inflation}_{it} + \beta_6 \log(\text{GDPpc}_{it}) \\ &+ \sum_y \gamma_y \text{Year}_{yt} + \sum_q \delta_q \text{Quarter}_{qt} + u_i + \varepsilon_{it}\end{aligned}$$

$$u_i \sim N(0, \sigma_u^2), \qquad \varepsilon_{it} \sim N(0, \sigma^2)$$

where:

- i indexes economies and t indexes quarters (2023 Q1 to 2025 Q4).
- Unemp_{it} is the unemployment rate (%) of economy i in quarter t.
- AI_{it} is the quarterly average share of online job postings requiring emerging AI skills (%).
- Adv_i equals 1 if economy i is classified by the IMF as an advanced economy and 0 if it is an emerging/developing economy (reference group).
- GDPgrowth_{it} is annual real GDP growth (%), Inflation_{it} is annual average CPI inflation (%), and GDPpc_{it} is GDP per capita in current U.S. dollars; these annual values are assigned to all quarters of the same year.
- Year_{yt} and Quarter_{qt} are indicator variables for calendar year (reference: 2023) and quarter (reference: Q1).
- β_0 is the intercept; β_1 is the association between AI adoption and log unemployment among emerging/developing economies; β_2 is the difference in log unemployment between advanced and emerging/developing economies when $\log(1 + \text{AI}) = 0$; β_3 is the difference in the AI slope between advanced and emerging/developing economies, so the AI slope among advanced economies is $\beta_1 + \beta_3$; β_4 to β_6 are associations with the macroeconomic covariates; γ_y and δ_q capture common year and seasonal effects.
- u_i is the random intercept for economy i, representing persistent differences in baseline unemployment across economies, and ε_{it} is the residual error.

Because the outcome is log-transformed, $100 \times (e^{\beta} - 1)$ gives the approximate percent change in the unemployment rate associated with a one-unit increase in a predictor, holding other variables constant.

4.2 Rationale for the statistical model

The outcome is continuous, and our goal is inference about the adjusted association between AI adoption and unemployment, so a linear regression framework is appropriate.

Our data have a repeated-measures structure: each economy contributes up to 12 quarterly observations, and observations from the same economy are likely to be correlated (e.g., South Africa has persistently high unemployment). Ordinary least squares would treat these observations as independent and understate standard errors. A random intercept for each economy accounts for this within-economy correlation. The interaction term tests whether the association between AI adoption and unemployment differs between advanced and emerging/developing economies, and we include macroeconomic covariates because Table 1 shows that the two groups differ systematically in GDP growth, inflation, and income, which may also affect unemployment. If mixed-effects models are not appropriate for this course, we will instead fit an ordinary linear regression with cluster-robust standard errors by economy.

4.3 Model Assumptions

1. **Linearity:** log unemployment is linearly related to each continuous predictor (on the scales used in the model).
2. **Independence:** after accounting for the economy-level random intercept, residuals are independent across observations. Residuals within the same economy may still be serially correlated across consecutive quarters, which we will check.
3. **Normality:** residuals ε_{it} and random intercepts u_i are approximately normally distributed.
4. **Equal variance:** residuals have constant variance across fitted values and across economy groups.
5. **No severe multicollinearity** among predictors.

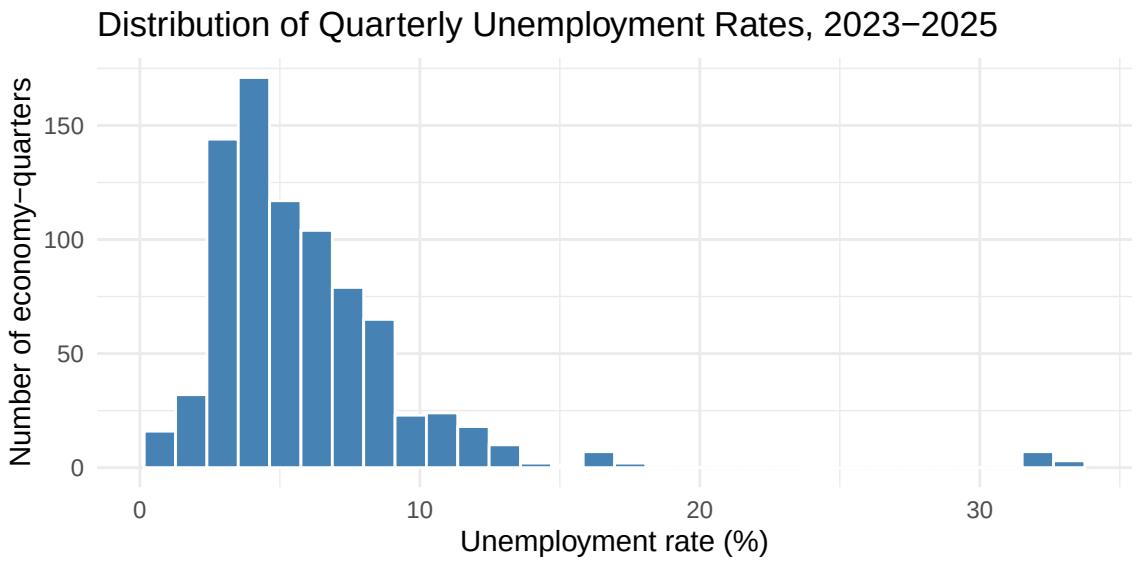
4.4 Model Assessment

- **Linearity and equal variance:** plot residuals against fitted values and against each continuous predictor, separately by economy group.
- **Normality:** Q-Q plots of the residuals and of the estimated random intercepts.
- **Independence:** examine the autocorrelation of residuals within economies across consecutive quarters.
- **Multicollinearity:** variance inflation factors (VIFs) for the fixed effects.
- **Influential observations:** identify influential economies by refitting the model leaving out one economy at a time and by examining influence diagnostics; we will pay particular attention to the small economies with very high AI shares and to extreme inflation values (e.g., Argentina), and will consider transforming or capping inflation if needed.

- **Interaction and model fit:** compare the models with and without the interaction term using a likelihood ratio test (fit by maximum likelihood), and report the intraclass correlation and marginal and conditional R^2.

Exploratory Data Analysis

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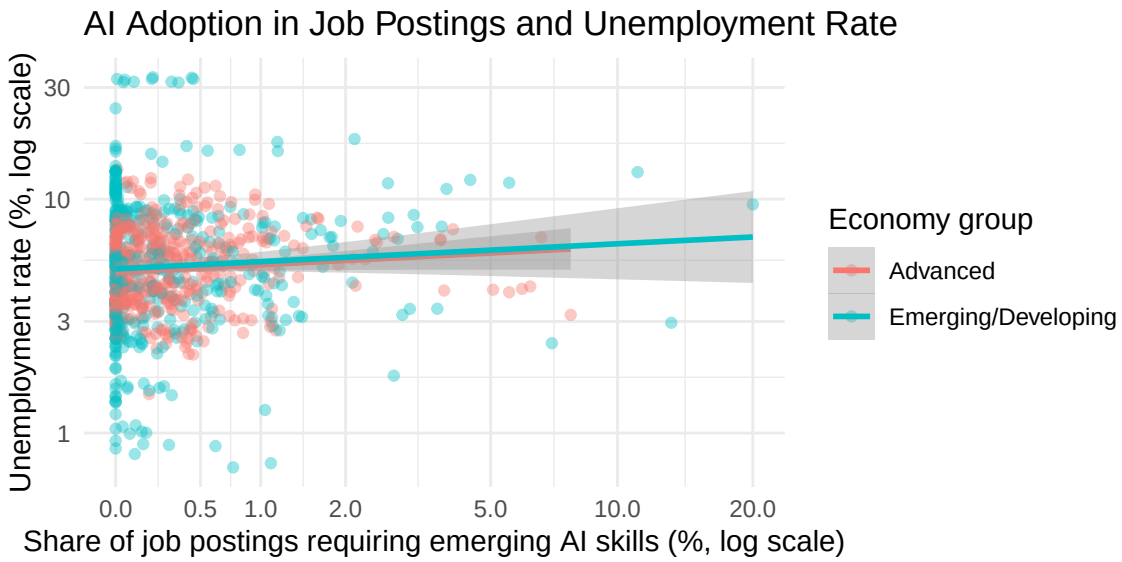


Table 1: Table 1. Median (IQR) of key variables by economy group

Characteristic	**Overall** N = 826	**Advanced** N = 403	**Emerging/Developing** N = 423
Unemployment rate (%)	5.17 (3.60, 7.39)	5.13 (3.73, 6.70)	5.26 (3.32, 8.33)
AI share, all occupations (%)	0.17 (0.01, 0.46)	0.22 (0.05, 0.50)	0.09 (0.00, 0.41)
AI share, high-exposure occupations (%)	0.21 (0.01, 0.59)	0.30 (0.07, 0.65)	0.12 (0.00, 0.52)
Missing	1	0	1
AI share, low-exposure occupations (%)	0.01 (0.00, 0.15)	0.06 (0.00, 0.17)	0.00 (0.00, 0.09)
Missing	16	3	13
Real GDP growth (%)	2.13 (0.90, 3.62)	1.48 (0.81, 2.22)	3.13 (1.49, 4.08)
Inflation (%)	3.73 (2.44, 5.91)	3.02 (2.32, 4.24)	4.68 (2.87, 8.68)
GDP per capita (USD)	23,576.27 (8,700.92, 48,976.04)	49,128.50 (34,376.84, 64,848.55)	9,749.71 (4,940.26, 15,914.25)

Table 1: Summary statistics for variables used in the analysis in Option 1

Outcome. The analytic sample contains 826 economy-quarter observations from 82 economies (403 from advanced economies and 423 from emerging/developing economies) between 2023 and 2025. The median quarterly unemployment rate is 5.17% (IQR: 3.60–7.39%). The distribution is right-skewed, with a small number of observations above 20% from South Africa and

Botswana, two economies with well-documented structural unemployment. We retain these observations and will consider a log transformation of the outcome.

Key relationship. The share of job postings requiring emerging AI skills is also highly right-skewed: most observations are below 2%, while a few smaller economies (e.g., Cyprus, Slovenia, Latvia, Bolivia, Bosnia and Herzegovina, and North Macedonia) exceed 5%. Because these economies have relatively small online job-posting markets, their high values may partly reflect measurement noise.

Covariates. Advanced economies have more than twice the median AI share of emerging/developing economies (0.22% vs. 0.09%), but they also differ systematically in GDP growth, inflation, and GDP per capita (Table 1). This highlights the need to adjust for these macroeconomic variables as potential confounders.

Dataset Option 2

Project Title: Remember Me: Does better sleep mean better cognition?

Data citation (use APA formatting or something similar): Health and Retirement Study. (2026). *RAND HRS longitudinal file 2022 (Version 2).* University of Michigan. https://hrsdata.isr.umich.edu/data-products/rand-hrs-longitudinal-file-2022

Original data collection date (when was the dataset published?): Oct 2026

Data pull date (when did you download the data you will use?): 4 Nov 2026

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1 Research Question

How do cognitive scores differ across high, medium, and low sleep quality, after accounting for age, education, and diabetes status, and does the association between sleep quality and cognitive performance vary by age?

2 Study Population

The observation unit is an individual respondent in the Health and Retirement Study (HRS) across 16 waves. The study is a longitudinal household survey conducted by the University of Michigan's Institute for Social Research and collects repeated survey information on health, cognition, employment, finances, and other aspects of aging in the United States.

For this analysis, we will focus on the data in the latest wave 16 in 2022.

3 Outcomes, Exposures, and Additional Variables of Interest

3.1 Primary Outcome(s)

Variable Name	Description	Coding specification
Cognitive Score	Cognitive Score combines multiple cognitive abilities testing including word recall, serial 7s, and backward counting, where higher scores indicate better performance.**0** = lowest possible cognitive performance on those tasks	0 - 27, where 0 = Lowest possible cognitive performance 27 = Highest possible cognitive performance

3.2 Primary Independent Variable(s)

Variable Name	Description	Coding specification
Sleep Quality	Sleep quality indicates how often respondents report feeling rested in the morning.	1 = Rarely/never 2 = Sometimes 3 = Most of the time

3.3 Additional Variables of Interest (including potential confounders)

Variable Name	Description	Coding specification
Age	Age is measured in years	Continuous numeric
Education	Respondent's years of education	Continuous numeric

4 Statistical Analysis Plan

4.1 Exploratory Data Analysis

Cognitive score is a continuous variable ranging from 0–27, so we will use a histogram to examine its distribution. Sleep quality is a categorical factor with three levels, so we will compare cognitive score distributions across sleep-quality groups using boxplots.

We will also examine the distributions of age and education and use summary statistics to describe the sample. For our hypothesis, we assume that respondents who report better sleep quality will have higher cognitive scores. It is also expected that this relationship may differ

by age, while accounting for education, since cognitive performance generally varies across age and educational background.

4.2 Statistical Model (write in mathematical notation and define each term)

$$\text{CognitiveScore}_i = \beta_0 + \beta_1 \text{SleepQuality}_i + \beta_2 \text{Age}_i + \beta_3 \text{Education}_i + \beta_4 (\text{SleepQuality}_i \times \text{Age}_i) + \epsilon_i$$

- CognitiveScore_i = Cognitive score for each respondents
- SleepQuality_i = Quality of sleep for each respondents
- Age_i = Respondent's age in years
- Education_i = Respondent's education level
- $\text{SleepQuality}_i \times \text{Age}_i$ = Interaction term testing whether the relationship between sleep quality and cognitive score differs by age
- ϵ_i = Error term

4.2 Rationale for the statistical model

We will use multiple linear regression because the outcome, cognitive score, is a numeric variable ranging from 0 to 27. This approach allows us to estimate differences in cognitive score across sleep-quality groups while also accounting for age and education.

4.3 Model Assumptions

- Linearity - The relationship between predicted value and the predictors is linear
- Independence - The observations are independent of one another.
- Normality - The residuals are approximately normally distributed.
- Equal variance - The residuals have approximately constant variance across predicted values.

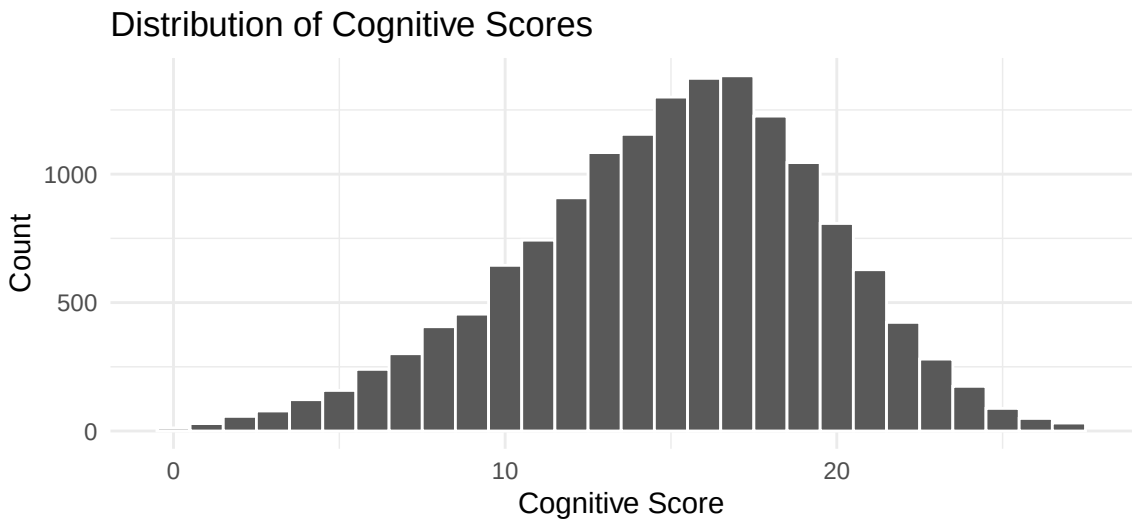
4.4 Model Assessment (how will you assess the assumptions and model performance?)

- Linearity - Assess the residuals vs. fitted plot for a random cloud with no clear pattern.
- Equal variance - Assess the residuals vs. fitted plot for a constant spread across the red line with no fanning behavior.
- Independence - Check that the residuals do not show obvious clustering or systematic patterns.
- Normality - Assess the Q-Q plot to see whether the residuals approximately follow 45 degree diagonal line

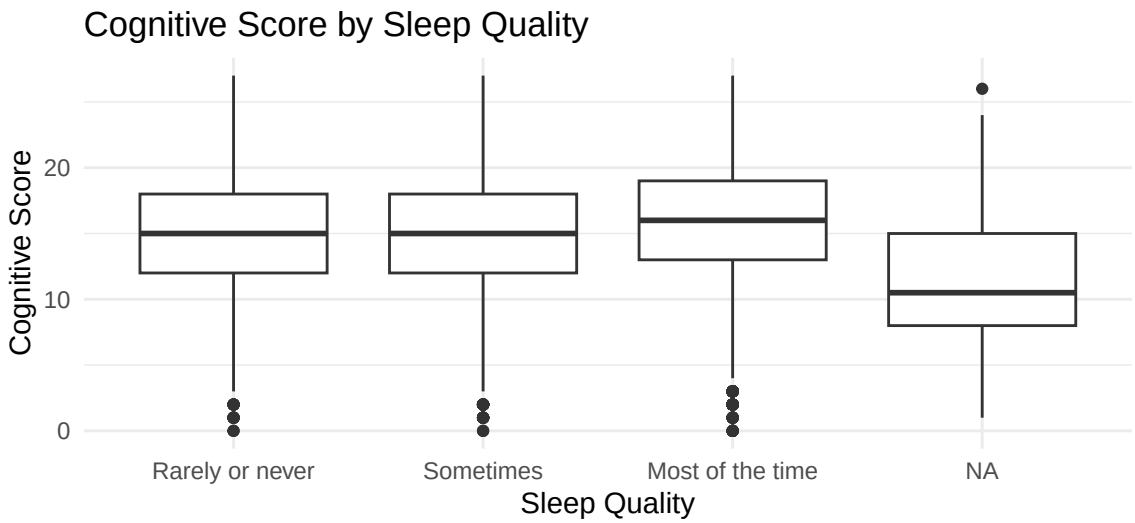
Exploratory Data Analysis

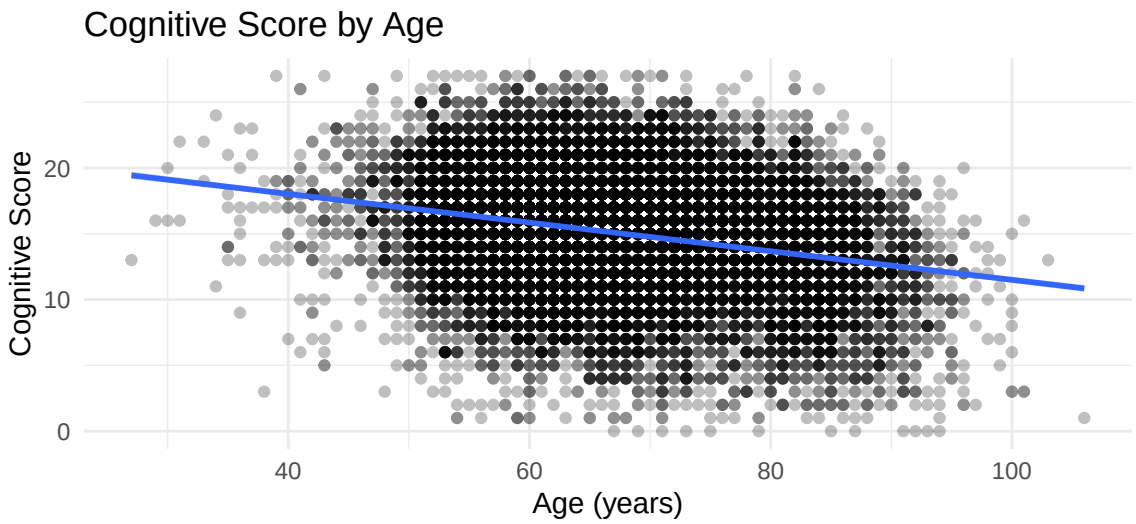
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$ raeduc     <fct> College and above, Less than high school, High school gradu~
```

Variable	Level	Count, N (%)	Mean (SD)
Cognitive score		15,201	15.05 (4.63)
Age (years)		15,787	67.73 (10.99)
Sleep quality	Rarely or never	2,692 (17.1%)	
	Sometimes	5,298 (33.7%)	
	Most of the time	7,719 (49.1%)	
Education level	Less than high school	11,210 (24.8%)	
	GED	2,234 (4.9%)	
	High school graduate	12,493 (27.7%)	
	Some college	10,241 (22.7%)	
	College and above	8,963 (19.9%)	




Distribution of Sleep Quality
30000
20000
10000
0
Count
Rarely or never
Sometimes
Most of the time
NA
Sleep Quality


Cognitive Score by Sleep Quality
20
10
0
Cognitive Score
Rarely or never
Sometimes
Most of the time
NA
Sleep Quality



Outcome: Cognitive scores range from 0 to 27, with a mean of 15.05 (SD = 4.63). The distribution is rough centered around a score of about 15, with relatively few observations at the lowest and highest values.

Key predictor: Among respondents with available sleep-quality information, 49.1% report feeling rested most of the time, 33.7% report sometimes feeling rested, and 17.1% report rarely or never feeling rested.

Key relationship

- Based on the box plot, cognitive scores appear to be slightly higher among respondents who report better sleep quality. Those who report feeling rested most of the time have a somewhat higher median cognitive score than those who report feeling rested sometimes or rarely/never.
- Scatter plot also demonstrates a clearer negative relationship with age. Overall, older respondents tend to have lower cognitive scores on average.

Missing Variables: Majority of the observations was missing because the survey tracked the respondents across 16 waves. Hence there were some individuals that did not partake in the latest wave. We will account for this when defining the analytic sample and report how many observations remain after restricting the analysis to respondents with complete data for the variables included in the model.

Team Charter

1. When will you meet as a team to work on the project components? Will these meetings be held in person or virtually? The meeting(s) will be held in person during weekdays and/or via Zoom as required outside of academic operational hours.

2. What is your group policy on missing team meetings (e.g., how much advance notice should be provided)? In the event of absence, the team member must communicate with the remainder of the team at least 24 hours in advance to reschedule accordingly.
3. How will your team communicate (email, Slack, text messages)? What is your policy on appropriate response time (within a certain number of hours? Nights/weekends?)? The team will communicate regularly through Slack during regular working hours (9:00am-5:00pm). Weekend communication is permissible during appropriate hours as well.